

Homework 3

BSTA 551

Directions

Same as previous instructions

For simulation problems, set your seed to `set.seed(551)` for reproducibility.

For the qmd of this Homework page, see this [github link](#)

You must show all of your work to receive credit.

This homework covers material from Lessons 6 and 7: Review of Point Estimation and Introduction to Confidence Intervals (pivotal quantities, two-sided z-intervals).

Questions

Problem 1: Pivotal Quantities and Confidence Intervals (Devore Section 8.1, Example 8.5 style)

A theoretical model suggests that the time-to-failure (in hours) of a certain medical device component follows an Exponential distribution with unknown rate parameter $\lambda > 0$. A random sample of $n = 8$ components yields the following failure times:

$$X = \{145, 89, 203, 67, 112, 156, 178, 94\}$$

Hint: If $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exponential}(\lambda)$, then $2\lambda \sum_{i=1}^n X_i \sim \chi_{2n}^2$ (chi-squared with $2n$ degrees of freedom).

(a) Let $h(X_1, \dots, X_n, \lambda) = 2\lambda \sum_{i=1}^n X_i$. Show that h is a pivotal quantity by verifying the definition included in class slides.

(b) Using the pivotal quantity from part (a), derive a $100(1 - \alpha)\%$ confidence interval for λ . Express your answer in terms of $\sum X_i$ and chi-squared critical values $\chi_{\alpha/2, 2n}^2$ and $\chi_{1-\alpha/2, 2n}^2$.

- (c) Compute a 95% confidence interval for λ using the data. In R, use `qchisq(p, df)` to find chi-squared quantiles.
- (d) Find a 95% confidence interval for the mean time-to-failure $\mu = 1/\lambda$.

Problem 2: z-Interval for a Normal Mean (Devore Section 8.1)

A pharmaceutical company is conducting quality control on the fill volume (in mL) of syringes. Based on extensive historical data, the fill volumes are known to follow a normal distribution with standard deviation $\sigma = 0.05$ mL. The target fill volume is $\mu_0 = 1.00$ mL.

A random sample of $n = 20$ syringes from a new production batch yields a sample mean of $\bar{x} = 0.982$ mL.

- (a) Verify that $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$ is a pivotal quantity for μ (verify the definition in the slides, you can just state the distribution of $\bar{X}$ to start, don't need to derive it mathematically).
- (b) Calculate a 95% confidence interval for the true mean fill volume μ using the observed sample mean provided.
- (c) Based on your confidence interval, is there evidence that the true mean fill volume differs from the target of 1.00 mL? Explain.
- (d) The company wants to estimate the mean fill volume to within ± 0.02 mL with 95% confidence. What sample size is required?

Problem 3: Devore Section 8.1, Exercise 1

Consider a normal population distribution with the value of σ known.

- (a) What is the confidence level for the interval $\bar{x} \pm 2.81 \cdot \sigma/\sqrt{n}$? (Hint, use the definition of critical value and the `pnorm()` function in R – do sanity checks to make sure you're using the appropriate tail!).
- (b) What is the confidence level for the interval $\bar{x} \pm 1.44 \cdot \sigma/\sqrt{n}$?
- (c) What value of $z_{\alpha/2}$ in the CI formula results in a confidence level of 99.7%?
- (d) Answer the question posed in part (c) for a confidence level of 75%.

Problem 4: Estimating Different Population Characteristics (Devore 7.4 Style)

A clinical researcher is studying verbal IQ scores in a population of patients recovering from mild traumatic brain injury. A random sample of $n = 12$ patients yields the following verbal IQ scores:

$$X = \{94, 108, 87, 115, 102, 99, 91, 106, 112, 98, 103, 89\}$$

Assume verbal IQ scores in this population follow a normal distribution with unknown mean μ and unknown variance σ^2 .

(a) Compute point estimates for:

- The population mean μ
- The population variance σ^2 (use the unbiased estimator S^2)
- The population standard deviation σ

(b) The researcher wants to estimate the proportion of patients in this population with verbal IQ scores below 95 (the clinical threshold for concern). If $X \sim N(\mu, \sigma^2)$, then this proportion is $p = P(X < 95) = \Phi\left(\frac{95 - \mu}{\sigma}\right)$.

Using the invariance property of MLEs and your estimates from part (a), compute a point estimate $\hat{p}$ for this proportion.

Hint: In R, use `pnorm()` to compute the standard normal CDF Φ .

(c) A colleague suggests a different estimator for p : simply use the sample proportion of observations below 95, i.e., $\tilde{p} = \frac{\#\{X_i < 95\}}{n}$.

- Compute $\tilde{p}$ for the given data
- Is $\tilde{p}$ unbiased for $p = P(X < 95)$? Explain briefly.
- (optional bonus question, do not need to complete for credit) Which estimator ($\hat{p}$ or $\tilde{p}$) do you think makes better use of the normality assumption? Why?

(d) Using R, compute the estimated standard error of $\bar{X}$:

$$\widehat{SE}(\bar{X}) = \frac{S}{\sqrt{n}}$$

If a different researcher collected a new sample of the same size, how much would you expect their sample mean to differ from yours, roughly speaking?

```
# Data
iq_scores <- c(94, 108, 87, 115, 102, 99, 91, 106, 112, 98, 103, 89)

# Your code here
```